Can an outburst help predict the next one?

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A century of observations, a small forecasting gain

SS Cygni is a pair of stars in which a white dwarf pulls gas from its companion into a disc. The disc flares irregularly, then fades. Amateur astronomers have followed these outbursts by eye for more than a century. Short and long flares tend to alternate. Earlier work found that longer flares are followed by longer quiet spells; our question was whether knowing a flare's length improves a forecast beyond what recent quiet spells already tell us.

We fitted two forecasters using 1910–1959 observations from the American Association of Variable Star Observers (AAVSO). Both knew the recent timing history; one also knew the completed flare's length. We froze the rules, published the protocol and then opened 1960–2017. The duration-aware forecast gained 0.0165 nats of log score per forecast (90% interval 0.0049 to 0.0281), over 441 forecasts: on a geometric average, about 1.7% more probability given to the outcomes as scored. This met the success rule set beforehand. The gain is small, but it is a held-out test, not a fit to known data.

The qualification matters. We scored probabilities assigned to intervals estimated by the flare-finding software. Those intervals are not known bounds on the true onset. The gain is therefore a result for this *working score*, not yet a clean measure of predicting the physical start. A formatting problem also required a documented reader repair after first access; it changed no observation or forecasting rule. The later years alone and the frozen long/short rule were inconclusive.

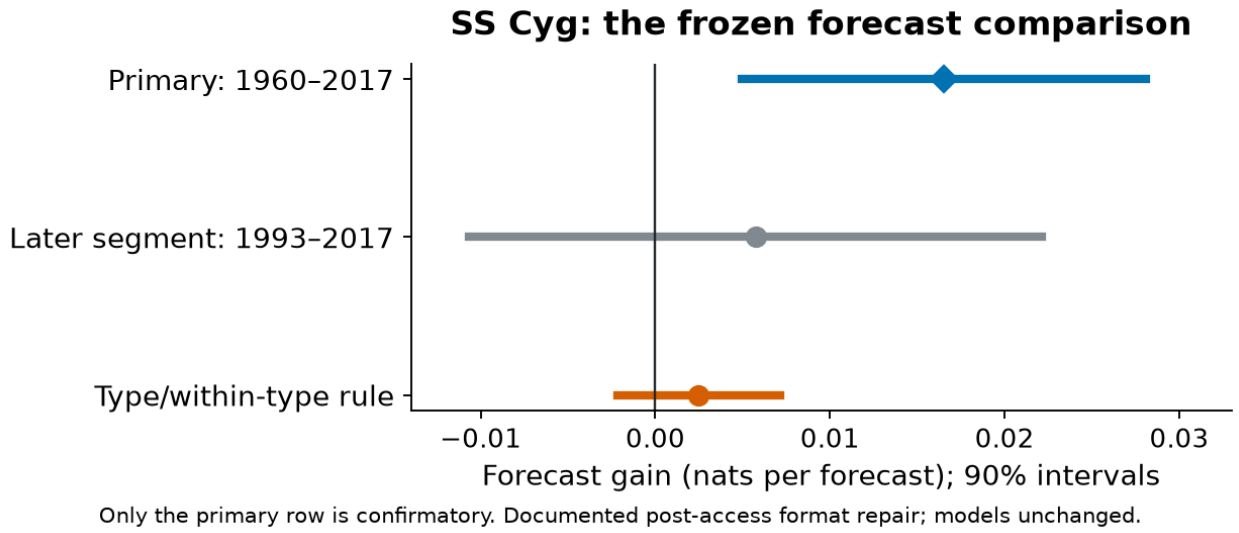


Figure. The original working-score gain and its interval are above zero. The later-period and long/short comparisons are inconclusive. None identifies physical memory.

What survived closer checking

The dates when people looked, the software's estimated start and end, and changes in the star's typical quiet interval can all affect the score. An error in the estimated end simultaneously makes one flare longer and the following quiet spell shorter. Observers may also watch harder when the star is active.

After external review we kept the frozen result and tested different measurements. Scoring at a single estimated start still favoured duration. New comparisons used report-based onset windows, different timing models and a clock starting at an observed faint report rather than an estimated end. Their duration gains were positive but their intervals included zero. Adjustments for changing typical gaps also reduced an analogous gain. These checks were designed after the original result: they expose its limits, and cannot count as fresh confirmation.

Could we test other stars?

We froze a feasibility screen before opening the early visual records of 10 other dwarf novae. None passed every rule for a later forecasting test. Under the same rules, SS Cyg's 1980s record gave 67 usable forecasts for 73 confirmed outbursts; the ten candidates together gave 102 for 485. Their largest loss was missing coverage of the previous quiet gap. Most of these stars are fainter than SS Cyg between outbursts, which makes continuous coverage by eye harder. A long bright-run flag excluded SS Aur; inspecting that flag found separated bright-report groups, not evidence for one continuous plateau or a particular cause.

The measured early-record rates fell well short of the planned sample. We stopped this expansion without fitting duration effects for these stars or opening their later test records. This says what the chosen rules could support, not that other dwarf novae lack the effect or that another archive or method could never work.

What we can say

Measured flare length improved the original frozen working score for SS Cyg, a test that rests on how closely amateur observers have watched this one star for a century. The observation model, baseline and measurement choices limit the inference. We have neither identified a physical memory mechanism nor demonstrated a general forecasting gain across dwarf novae. SS Cyg's records from 2018 onward and the candidates' records from 1990 onward remain closed.

How this work was done

Barry Brook set the direction and access boundaries and decides release. Claude reviewed the literature, built the flare finder, drafted the first versions and reviewed these. Codex built and ran the forecasts, independently checked the results and later diagnostics, and prepared these second drafts. GPT-6 Pro supplied the initial ideas and an external review. The public repository preserves the frozen calculation, corrections, failed transfer and evidence behind each claim. These are second drafts for review.

Sources

Cannizzo and Mattei (1992) analysed the historical duration–gap relation; Cannizzo (1993) modelled the disc cycle. The protocol is archived at https://doi.org/10.5281/zenodo.24053338.

Cannizzo, John K. 1993. "The Accretion Disk Limit Cycle Model: Toward an Understanding of the Long-Term Behavior of SS Cygni." *The Astrophysical Journal* 419: 318. https://doi.org/10.1086/173486.

Cannizzo, John K., and Janet A. Mattei. 1992. "On the Long-Term Behavior of SS Cygni." *The Astrophysical Journal* 401: 642. https://doi.org/10.1086/172092.